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Seminar Paper

Reverse Vending Machine Choice and Deployment under Discrete Choice Models

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Abstract

Reverse vending machines (RVMs) collect used beverage containers for recycling. Where an operator should install these machines depends on how users choose between them, and the two questions are rarely studied together. We analyse 2,255 return decisions made by 1,830 users of an RVM network in Medellín, Colombia. A multinomial logit model estimates the probability that a user returns containers at one of 35 candidate machines or does not return at all. Location type has the strongest influence on this choice: machines at public and private buildings are clearly preferred. The deterrent effect of distance is present for outdoor machines and absent for indoor machines. The estimated probabilities then enter a mixed-integer program that uses an exact linear reformulation of the logit probabilities and selects the subset of candidate locations maximising the expected profit from the collected material under a budget constraint. In our experiments the model installs a single machine, which captures 90–95% of the choice probability. A budget sensitivity analysis shows that additional funds leave this solution unchanged; how many machines are worth installing is determined by the economic attractiveness of the candidate sites relative to their fixed costs.

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1. Introduction

Packaging is the most common application of plastic, and single-use beverage containers make up a growing share of it [Geyer et al., 2017]. Plastic is popular as a packaging material because it is durable and cheap to produce [Somarelli et al., 2026]. Its disposal, however, is a problem. In 2019, only about 9% of the plastic waste generated worldwide was recycled, while roughly 22% was mismanaged; an estimated 22 megatonnes leaked into the natural environment in that year alone [OECD, 2022].

Macroplastics such as bottles, bags and discarded fishing nets entangle several hundred animal species, which often leads to injury or death [Savoca et al., 2026]. Smaller micro- and nanoplastics (MNPs) are ingested by hundreds of species and have also been detected in indoor air. In humans, MNPs have been found in blood, heart and brain tissue. Exposure has been associated with a higher risk of dementia and of certain cancers, although the causal evidence for these associations is still limited [Somarelli et al., 2026]. Plastic leakage therefore concerns public health as well as ecosystems.

One policy response is *extended producer responsibility* (EPR), which makes producers responsible for their products over the entire life cycle, including the stage after consumption. The responsibility can be physical or financial, and it can be transferred to the producers fully or in part [OECD, 2016]. A common instrument to implement EPR is the *deposit-refund system* (DRS): consumers pay a small deposit when they buy a beverage and get it back when they return the empty container to a designated collection point. The DRS follows the idea of a circular economy, which aims to keep materials in use at the highest possible quality for as long as possible [Geissdoerfer et al., 2017].

Reverse vending machines (RVMs) are the machines at which consumers return their containers at such collection points. An RVM detects which type of container is inserted and, under a DRS, pays out the corresponding refund. Operators usually install RVMs at busy locations such as supermarkets, parks or universities. So far, little research exists on how consumers choose between the available machines and on where new machines should be placed. A recent bibliometric review counts only

a handful of studies on RVMs in the circular economy [Fernandes and Maurício C., 2025]. For related planning problems, however, there is an established stream of research on facility location under customer choice: Benati and Hansen [2002] formulate the maximum capture problem, in which customers choose between facilities according to a multinomial logit model, and Haase and Müller [2014] compare linear reformulations of this problem that standard solvers can handle. We build on this work and transfer it to the placement of RVMs.

This paper develops a discrete choice model that estimates the probability that a consumer returns a beverage container at a given RVM, or does not return it at all. The probability depends on characteristics of the user and on attributes of the machines, such as their distance and the type of location where they stand. The estimated probabilities then enter a mixed-integer program that selects, from a set of candidate locations, the subset of RVMs that maximises the expected volume of collected plastic. The model covers two connected decisions: which machine a user chooses, and how much material the operator can expect to collect at the chosen sites.

Chapter 2 reviews the literature on discrete choice modelling and on choice-based facility location, and positions this paper within it. Chapter 3 describes the data from Medellín and the structure of the observed choices. Chapter 4 develops the discrete choice model and reports its estimates. Chapter 5 embeds the estimated probabilities in the deployment model, and Chapter 6 presents the computational results. Chapter 7 concludes and outlines future steps.

2. Literature Review

This chapter places the project within the existing literature. Section 2.1 discusses how the discrete choice model of user behaviour has been combined with facility location decisions in prior work, and how the resulting computational problem has been addressed. Section 2.2 turns to work closely related to RVM deployment specifically. Two papers there, together, define almost exactly the gap this project fills. Section 2.3 draws everything together into the project's positioning. The discrete choice model itself, together with the behavioural assumptions it rests on, is introduced in Chapter 4.

2.1 Choice-Based Facility Location

Once user choice is modelled probabilistically, a natural planning question follows: which subset of candidate facilities should a planner open to maximise captured demand? This is the choice-based facility location problem. Benati and Hansen [2002] first formalised it and showed it to be NP-hard. Making the set of open facilities a decision variable turns the multinomial logit (MNL) probability itself into a function of that decision. This function is nonlinear, and this nonlinearity is the central computational difficulty the subsequent literature addresses.

Haase and Müller [2014] resolve this using precomputed attractiveness ratios between each facility and the opt-out alternative. Precisely because of the independence of irrelevant alternatives (IIA) property of the MNL, these ratios let the nonlinear probability be replaced by an equivalent system of linear constraints. This reformulation is exact for the MNL. It is also computationally tractable with standard solvers, and it is the formulation this project builds on directly in Chapter 5. For larger instances, the size of this reformulation becomes a bottleneck. Ljubić and Moreno [2018] and Mai and Lodi [2020] address this with outer-approximation methods. Rather than the full linearisation, these methods approximate the objective iteratively. This solves instances far beyond the reach of the exact formulation. They are not needed at the instance size considered in this project, but are noted here for completeness. They define the broader solution landscape this project's formulation sits within.

2.2 Related Applications

Two papers are particularly relevant to this project, for different reasons. One applies the same methodology to a structurally similar problem. The other studies RVM location directly, but without a behavioural demand model.

Lin et al. [2020] apply the Haase and Müller [2014] formulation to parcel locker placement. There, customers choose between available lockers and home delivery. Home delivery is an opt-out alternative directly analogous to not returning a container in the RVM setting. Their case study confirms that this opt-out alternative matters empirically: models that omit it systematically overestimate captured demand. This is relevant to how the opt-out alternative is specified in Chapter 4.

Bedoya et al. [2024] study RVM location specifically, and in a setting close to this project's own: 35 candidate sites in Medellín, operated by a Colombian recycling startup. Their model is multiobjective. It maximises user visits and collected material while minimising installation cost, subject to a minimum-distance constraint between machines to avoid locating them too close together. Demand at each site, however, is treated as a known input, estimated from historical visit and collection data. It is not the outcome of a probabilistic choice process. Notably, their own concluding discussion identifies this as a limitation. They explicitly suggest embedding discrete choice models "following the approach presented in Haase and Müller (2014)" as a direction for future work – the exact methodology this project applies.

A third, very recent paper further confirms this gap. Gomes da Silva et al. [2026] study RVM location using spatial interaction models in the tradition of Huff [Huff, 1962, 1964]. There, each facility's choice probability is proportional to its attractiveness and inversely related to distance. Their approach incorporates an explicit demand function and multiple sustainability objectives – social, environmental, and economic. Echoing Bedoya et al. [2024], the authors note that theirs is only the second modelling-based approach to RVM location in the literature. Unlike the MNL, however, the Huff-style probability is not derived from an estimated random utility model with statistically tested parameters. Attractiveness is specified rather than estimated from observed choices. This project's use of an MNL estimated by maximum likelihood on observed choice data (Chapter 4) therefore remains, to our knowledge, the first behaviourally estimated, choice-based approach to RVM location specifically.

A fourth connection concerns the behavioural assumption itself, rather than RVMs directly. Rogosinski et al. [2026] study a setting where the IIA property underlying the MNL is deliberately relaxed. They use a mixed logit model with randomly distributed taste parameters across customers, applied to assortment optimisation rather than facility location. A widely used approximation for this problem ranks alternatives by revenue. They show that this approximation can perform arbitrarily badly once the model allows correlated substitution patterns between alternatives. This is precisely the behaviour that the MNL's IIA property rules out by assumption. Their result does not change how the MNL is used in this project. But it clarifies what is being assumed away: whenever locations compete for the same nearby users in ways not captured by measured attributes, the MNL will not represent that competition correctly. This is picked up again in Chapter 7.

2.3 Research Gap and Project Positioning

Sections 2.1 and 2.2 together outline the gap this project addresses. A tractable, exact methodology for choice-based facility location exists [Haase and Müller, 2014]. It has been applied successfully to at least one closely related domain [Lin et al., 2020]. RVM location, meanwhile, has been studied using deterministic covering models [Bedoya et al., 2024] and, more recently, gravity-based spatial interaction models [Gomes da Silva et al., 2026] – but not using a discrete choice model estimated from observed user behaviour. Bedoya et al. [2024] flag this omission themselves as future work. This project closes that gap directly: an MNL is estimated from empirical RVM choice data (Chapter 4), and its estimated preferences enter the Haase and Müller [2014] formulation to determine optimal locations (Chapter 5).

3. Data

This chapter describes the data set behind the choice and deployment models. We first introduce the study context and the reverse vending machines (RVMs) it covers, then explain how a single observation is structured, which preparation steps we applied, and which attributes are available for modelling.

3.1 Study Context and Data Source

The data were collected in Medellín, Colombia. They describe how individual users return beverage containers at a network of RVMs across the city. Each RVM is a physical collection point at a specific location, for example in a shopping centre, a public park or on a university campus. For every recorded return event the data contain who returned containers, which machine the person chose, how far that machine was from the user, and how much material was returned. The exact collection period is not documented in the material available to us.

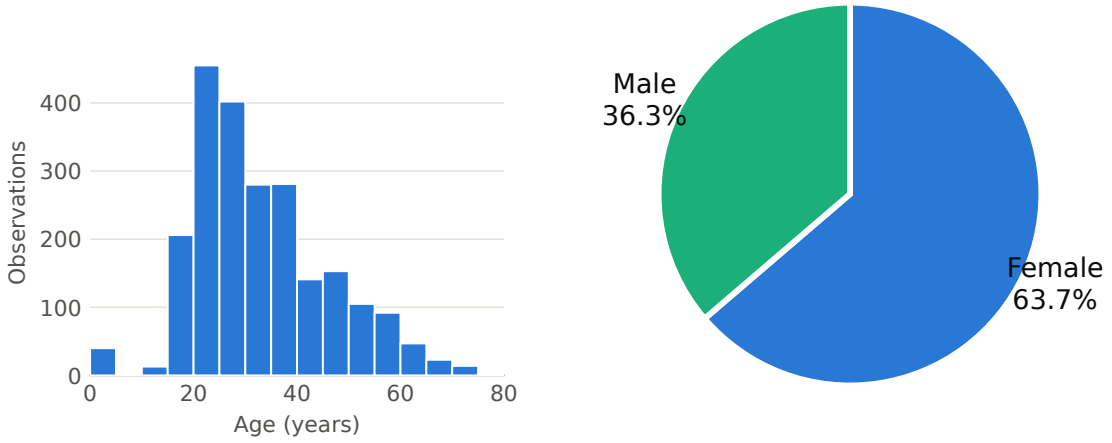
3.2 Observations and Choice Structure

The data set contains 2,255 observations from 1,830 distinct users. Each observation is one choice situation: a user faces a common set of 35 candidate RVM locations and either returns containers at exactly one of them or does not return at all. The chosen alternative is stored in the variable `choice_code`, with values from 1 to 36. Codes 1–35 identify the selected machine and code 36 stands for the no-return option, which occurs in 39 observations. For every choice situation the data provide attributes of the user and attributes of each alternative, so both can enter the utility of an option in a discrete choice model.

The data required little preparation. The only transformation we applied is a rescaling of `Age`, which we divide by ten so that its coefficient is easier to interpret and lies in a similar range as the other estimates. All remaining variables enter the models as they appear in the original file.

User-specific attributes.

These describe the decision maker and do not vary across the alternatives of a choice situation. They include the user's `Age` (mean 32.6 years, median 30), a `Female` indicator (about 63.7% of observations), a `socioeconomic_level` on a scale from 1 to 6, the geographic coordinates of the user (`Latitud`, `Longitud`), the number of return visits (`number_of_visits_per_user`) and the total volume returned (`volume_gemerated_per_user_grams`, in grams). Figure 3.1 shows the age distribution and the gender split of the observations.



(a) Age distribution. **(b)** Gender split.

Figure 3.1: Demographics of the observed choice situations ($n = 2{,}255$). Most users are between 20 and 40 years old, and about two thirds are female. Age values of zero (37 observations) most likely indicate missing entries.

Alternative-specific attributes.

These vary across the 35 candidate machines and are stored in wide format, with one column per attribute and alternative. The data provide the distance between the user and every machine, an indoor indicator for each machine, a location type indicator for every machine and indicators for the socioeconomic level of the machine's neighbourhood. With 35 alternatives and 19 attributes each, this block explains the large number of columns in the raw file.

Outcome variables.

Two variables measure the intensity of the return decision and complement the discrete machine choice: `number_of_visits_per_user` (mean 8.9, median 3) and `volume_gemerated_per_user_grams` (mean 155.8 g, median 57 g). In both cases the mean is far above the median, so a small group of users accounts for much of the returned volume.

4. Discrete Choice Model

A discrete choice model estimates the probability that an individual selects a particular reverse vending machine from a set of available alternatives. These estimated choice probabilities are then used as input for the mixed-integer programming model presented in Chapter 5.

4.1 Core Idea

Discrete choice models assume that each individual associates a utility with every available alternative and chooses the one with the highest utility. Let N denote the set of individuals and I the set of available machines. The utility that individual $n \in N$ gets from choosing machine $i \in I$ is written as [Train, 2009]

$$U_{ni} = V_{ni} + \varepsilon_{ni}, \tag{4.1}$$

where V_{ni} is the observed part of utility, and ε_{ni} is the unobserved part.

The observed utility V_{ni} is specified as a function of observable characteristics of the individual and the machine. In this project, these characteristics include variables such as distance, machine location type, indoor status, age and gender. The unobserved term ε_{ni} captures factors that affect the individual's decision but are not included in the dataset, such as personal preferences or previous experience.

The person chooses machine i when its utility is greater than the utility of every other available alternative:

$$U_{ni} \geq U_{nj} \qquad \forall j \in C_n, \tag{4.2}$$

where C_n denotes the choice set available to individual n.

Because the researcher does not observe the error terms, the choice cannot be predicted with certainty. Instead, the model calculates the probability that each alternative provides the highest utility. The unobserved utility terms are assumed to be independently and identically distributed according to the Gumbel distribution.

This assumption leads to the logit model [Train, 2009], for which the probability that individual n chooses machine i is

$$P_{ni} = \frac{\exp(V_{ni})}{\sum_{j \in C_n} \exp(V_{nj})}. \tag{4.3}$$

This closed-form choice probability defines the multinomial logit model (MNL), the standard discrete choice model in applied work [McFadden, 1974; Train, 2009], and its parameters are estimable by maximum likelihood. A property central to this project is the independence of irrelevant alternatives (IIA). It states that the relative odds between any two alternatives do not depend on which other alternatives are available. This is convenient for estimation and, as discussed in Chapter 5, for optimisation. But it is also a simplifying assumption. Its consequences resurface in Chapter 7 and in Section 2.2.

4.2 Utility Specification

The specification of the utility function is an important step in the development of the discrete choice model. The estimated utilities determine the choice probabilities and directly affect the input provided to the optimisation model. Rogosinski et al. [2026] emphasise that the performance of an optimisation approach can depend on the specification of customer utility.

Because of that, several utility specifications were tested during the model development. The specifications were evaluated based on statistical significance, numerical stability, and their suitability for use in the optimisation model. Rather than presenting every specification, we focus more on three specific models.

The discrete choice models were estimated using Biogeme, a Python package designed for maximum likelihood estimation of discrete choice models [Bierlaire, 2023].

4.2.1 Model 1: Specification with All 35 Machines

Although this is not the first model estimated during the project, it represents an important stage in understanding the choice-set definition. In this specification, all 35 candidate machines are included as available alternatives for every person.

Most location variables are grouped into broader categories. Public and private buildings are combined into the variable *building*, while universities, technical institutions, and science parks are combined into the variable *campus*. Industrial business centres are represented by the variable *business*.

The grouping reduces the number of coefficients and avoids unstable estimates for rarely chosen location types. This is important because the optimisation model later exponentiates utilities. Public parks, hospitals, recreational parks, downtown recycling locations, and convention centres are not included because they are rarely chosen or do not provide reliable estimates in earlier models.

The utility of machine i for individual n is specified as

$$\begin{aligned} V_{ni} = {} & \beta_{\text{dist}}\, dist_{ni} + \beta_{\text{indoor}}\, indoor_i + \beta_{\text{dist_indoor}}\, dist_{ni} indoor_i \\ & + \beta_{\text{indoor_age}}\, indoor_i Age10_n + \beta_{\text{indoor_female}}\, indoor_i Female_n \\ & + \beta_{\text{building}}\, building_i + \beta_{\text{business}}\, business_i. \end{aligned} \tag{4.4}$$

The opt-out alternative is represented by an alternative-specific constant:

$$V_{n0} = ASC_{\text{out}}. \tag{4.5}$$

Table 4.1 reports the fit of this specification, and Table 4.2 the estimated coefficients.

Table 4.1: Estimation results for the 35-machine specification.

Measure	Value
Number of observations	2255
Number of estimated parameters	8
Final log likelihood	−6094.668
AIC	12205.34
BIC	12251.10
Smallest eigenvalue	5.956
Condition number	2456.68

Table 4.2: Estimated coefficients for the 35-machine specification.

Parameter	Estimate	Robust SE	Robust t-stat.	Robust p-value
ASC_{OUT}	−1.07	0.171	−6.24	4.41×10^{-10}
β_{DIST}	−0.158	0.0121	−13.1	0
β_{INDOOR}	−1.11	0.246	−4.52	6.05×10^{-6}
$\beta_{\text{DIST_INDOOR}}$	0.0683	0.0156	4.37	1.24×10^{-5}
$\beta_{\text{INDOOR_AGE}}$	−0.342	0.0730	−4.69	2.80×10^{-6}
$\beta_{\text{INDOOR_FEMALE}}$	0.369	0.173	2.14	0.0326
β_{BUILDING}	1.88	0.0439	42.8	0
β_{BUSINESS}	−2.19	0.410	−5.35	9.01×10^{-8}

Assuming that all 35 machines belong to the choice set of every individual is the main limitation of this specification. This would imply that each person considers all 35 machines as a realistic alternative. In practice, this assumption is unlikely, since respondents may only know about machines located near their home, workplace or usual travel routes.

The choice set is restricted to the 12 machines that appear as observed choices in the variable `choice_code`, together with the opt-out alternative.

The coefficients in the utility function are generic attribute coefficients, meaning that they represent the effects of observed characteristics such as distance and location type rather than the effect of a particular machine [Train, 2009]. Since the same attributes are available for all candidate machines, the estimated utility function can then be evaluated for all 35 machines considered in the optimisation model.

4.2.2 Models 2 and 3: Specifications with Only Chosen Machines

The second and third models are estimated using the 12 machines that appear as observed choices in the variable `choice_code`, together with the opt-out alternative.

These two specifications use the same observations and the same choice set, which makes their estimation results directly comparable.

The two models differ only in the treatment of the `indoor` variable. Model 2 includes an indoor coefficient, while Model 3 removes it.

The utility in Model 2 is specified as

$$\begin{aligned} V_{ni}^{(2)} = {} & \beta_{\text{dist}}\, dist_{ni} + \beta_{\text{indoor}}\, indoor_i + \beta_{\text{dist_indoor}}\, dist_{ni} indoor_i \\ & + \beta_{\text{indoor_age}}\, indoor_i Age10_n + \beta_{\text{indoor_female}}\, indoor_i Female_n \\ & + \beta_{\text{building}}\, building_i + \beta_{\text{business}}\, business_i + \beta_{\text{campus}}\, campus_i. \end{aligned} \tag{4.6}$$

The utility in Model 3 is specified as

$$\begin{aligned} V_{ni}^{(3)} = {} & \beta_{\text{dist}}\, dist_{ni} + \beta_{\text{dist_indoor}}\, dist_{ni} indoor_i \\ & + \beta_{\text{indoor_age}}\, indoor_i Age10_n + \beta_{\text{indoor_female}}\, indoor_i Female_n \\ & + \beta_{\text{building}}\, building_i + \beta_{\text{business}}\, business_i + \beta_{\text{campus}}\, campus_i. \end{aligned} \tag{4.7}$$

For both specifications, the opt-out utility is given by

$$V_{n0} = ASC_{\text{out}}. \tag{4.8}$$

Table 4.3: Comparison of Models 2 and 3.

Measure	**Model 2**	**Model 3**
Choice set	12 machines + opt-out	12 machines + opt-out
Number of observations	2255	2255
Number of estimated parameters	9	8
Final log likelihood	−4579.024	−4579.035
AIC	9176.048	9174.071
BIC	9227.536	9219.838
Smallest eigenvalue	5.932	5.935
Condition number	1601.14	1592.94

Table 4.3 compares the fit of the two models. Model 3 is obtained from Model 2 by setting the indoor coefficient to zero. Following Train [2009, Section 3.8.2] the likelihood-ratio statistic is

$$LR = 2\,(LL_{\text{unrestricted}} - LL_{\text{restricted}}). \tag{4.9}$$

Using the final log-likelihood values gives

$$LR = 2\,[-4579.024 - (-4579.035)] = 0.022. \tag{4.10}$$

The models differ by one parameter, so the test has one degree of freedom. At the 5% significance level, the chi-square value is 3.841. Since $0.022 < 3.841$, we do not

Table 4.4: Estimated coefficients for Model 3.

Parameter	Estimate	Robust SE	Robust t-stat.	Robust p-value
ASC_{OUT}	-1.67	0.180	-9.28	< 0.001
β_{DIST}	-0.116	0.0148	-7.83	< 0.001
$\beta_{\text{DIST_INDOOR}}$	0.123	0.0267	4.62	< 0.001
$\beta_{\text{INDOOR_AGE}}$	-0.330	0.0545	-6.04	< 0.001
$\beta_{\text{INDOOR_FEMALE}}$	0.368	0.165	2.23	0.026
β_{BUILDING}	1.58	0.0535	29.6	< 0.001
β_{BUSINESS}	-2.89	0.411	-7.03	< 0.001
β_{CAMPUS}	-0.305	0.0717	-4.26	< 0.001

reject the null hypothesis that the indoor coefficient is equal to zero. Model 3 is then chosen because it is simpler and gives almost the same fit.

Table 4.4 shows that all retained coefficients are statistically significant at the 5% level. Distance has a negative effect on utility, while the positive distance-indoor interaction indicates that this distance penalty is weaker for indoor machines. The utility of indoor machines decreases with age but for female respondents it is higher. Building locations have a strong positive effect, while business and campus locations have negative effects relative to the reference category. The negative and significant opt-out constant indicates that choosing no machine has lower baseline utility than choosing an RVM, all else equal.

4.2.3 Elasticity of Model 3

The elasticity with respect to distance measures the percentage change in the choice probability caused by a one-percent change in distance [Cordera et al., 2022]. Since Model 3 includes an interaction between distance and indoor status, the direct distance elasticity is

$$E_{ni}^{\text{dist}} = (\beta_{\text{dist}} + \beta_{\text{dist_indoor}} indoor_i)\, dist_{ni}(1 - P_{ni}). \tag{4.11}$$

With

$$\beta_{\text{dist}} + \beta_{\text{dist_indoor}} = -0.116 + 0.123 = 0.007. \tag{4.12}$$

Since the combined effect is positive but also very close to zero, it is important to test if it is statistically different from zero [Koppelman and Bhat, 2006, Section 5.7.3.1]. Because the indoor distance effect is the sum of two estimated coefficients, its standard error is calculated using

$$\operatorname{Var}\left(\hat{\beta}_{\text{dist}} + \hat{\beta}_{\text{dist_indoor}}\right) = SE_{\text{dist}}^2 + SE_{\text{dist_indoor}}^2 + 2\operatorname{Cov}\left(\hat{\beta}_{\text{dist}}, \hat{\beta}_{\text{dist_indoor}}\right). \tag{4.13}$$

Using the estimates from Model 3 gives a standard error of approximately 0.0279 and the test statistic is

$$t = \frac{0.007}{0.0279} \approx 0.25. \tag{4.14}$$

Because of the absolute value of the test statistic being below 1.96, the combined distance effect for indoor machines is not statistically different from zero at the 5% significance level. That is why, the positive estimate of 0.007 should not be interpreted as evidence that greater distance increases the probability of choosing an indoor machine.

5. Deployment Model

After deriving the user choice probabilities from the Multinomial Logit (MNL) model, the next step is to address the optimisation component of the reverse vending machine facility location problem.

While the MNL model captures users' decision-making behaviour and estimates the probability of choosing a particular location, it does not provide any information about which set of locations should be selected to achieve an optimal system configuration.

To determine the optimal locations from the predefined set of candidate reverse vending machine (RVM) sites, we therefore employ a Mixed Integer Programming (MIP) approach. In our problem, two different types of decisions have to be considered. The choice probabilities obtained from the MNL model are continuous variables, whereas the decision of whether or not to install an RVM at a specific candidate location is binary and therefore discrete.

A Mixed-integer Programming formulation enables both continuous and discrete decision variables to be incorporated into a single optimisation framework. This makes it particularly well suited for the facility location problem considered in this study. Furthermore, a mixed-integer linear programming formulation offers a systematic framework for solving complex optimisation problems and can be solved efficiently using standard optimisation software [Krohn et al., 2021].

5.1 Problem Formulation and Motivation

The optimisation model is based on the choice probabilities estimated by the discrete choice model. To incorporate the facility location decisions into the optimisation model, a linear reformulation is required. The MNL model provides the choice probabilities in the following form:

$$P_{ni} = \frac{\exp(V_{ni})}{\exp(V_{\text{out}}) + \sum_{j \in I} \exp(V_{nj})} \tag{5.1}$$

The key idea of the optimisation part is to introduce a binary decision variable y_i, where $y_i = 1$ if an RVM is installed at candidate location i, $y_i = 0$ otherwise, since not every location is going to be installed in an optimal solution. By introducing the binary variable, the choice probability expression changes to:

$$P_{ni} = \frac{\exp(V_{ni})\, y_i}{\exp(V_{\text{out}}) + \sum_{j \in I} \exp(V_{nj})\, y_j} \tag{5.2}$$

Since the decision variables appear in the denominator, this equation is no longer linear, which is why a linear reformulation is needed [Haase and Müller, 2014].

5.2 Mathematical Formulation

Let N denote the set of users or observations, and let I denote the set of candidate RVM machines. The estimated opt-out utility is $V_{\text{out}} = ASC_{\text{out}} = -1.67$, and the MIP uses the normalised utility

$$\tilde{V}_{ni} = V_{ni} - V_{\text{out}}. \tag{5.3}$$

Based on the normalised utility $\tilde{V}_{ni}$, we define two constants that are used throughout the MIP formulation:

$$\kappa_{ni} = \exp\left(\tilde{V}_{ni}\right) = \exp(V_{ni} - V_{\text{out}}) \tag{5.4}$$

and

$$\omega_{ni} = \frac{\exp\left(\tilde{V}_{ni}\right)}{1 + \exp\left(\tilde{V}_{ni}\right)} \tag{5.5}$$

κ_{ni} is an attractiveness parameter, representing the relative ratio of the attractiveness of machine i to that of the opt-out alternative. ω_{ni} serves as an upper bound for the choice probability of user n to choose machine i. This constant primarily ensures that the probability does not grow unrealistically large and is consistent with the probabilities derived from the discrete choice model.

We further introduce the following variables:

$$y_i = \begin{cases} 1, & \text{if candidate location } i \text{ is selected,} \\ 0, & \text{otherwise.} \end{cases} \tag{5.6}$$

$$X_{ni} \geq 0 \tag{5.7}$$

$$\tilde{X}_n \geq 0 \tag{5.8}$$

As already introduced above, the binary decision variable y_i takes value 1 if a candidate RVM location is installed, and 0 otherwise. X_{ni} denotes the probability of user n to choose machine i. Finally we consider the probability of the opt-out alternative (user n chooses no available machine) $\tilde{X}_n$.

The objective is to maximise the profit generated by the amount of collected plastic volume.

$$\max \quad Z = R \sum_{n \in N} g_n \sum_{i \in I} X_{ni} - \sum_{i \in I} F_i y_i \tag{5.9}$$

where g_n is the amount of collected plastic volume by user n, R is the monetary value per gram of g_n. F_i denotes the fixed cost of installing machine i.

The objective function is subject to the following constraints:

$$\tilde{X}_n + \sum_{i \in I} X_{ni} = 1, \qquad \forall n \in N. \tag{5.10}$$

$$X_{ni} \leq \omega_{ni} y_i, \qquad \forall n \in N,\ i \in I. \tag{5.11}$$

$$X_{ni} \leq \kappa_{ni} \tilde{X}_n, \qquad \forall n \in N,\ i \in I. \tag{5.12}$$

$$\tilde{X}_n - \frac{1}{\kappa_{ni}} X_{ni} + y_i \leq 1, \qquad \forall n \in N,\ i \in I. \tag{5.13}$$

$$\sum_{i \in I} y_i \leq K. \tag{5.14}$$

$$\sum_{i \in I} F_i y_i \leq B. \tag{5.15}$$

$$y_i \in \{0, 1\}, \qquad \forall i \in I. \tag{5.16}$$

$$X_{ni} \geq 0, \qquad \forall n \in N,\ i \in I. \tag{5.17}$$

$$\tilde{X}_n \geq 0, \qquad \forall n \in N. \tag{5.18}$$

The above constraints are derived as follows: To ensure that user n chooses either one of the available RVM locations or the opt-out alternative, all probabilities have to sum up to 1 (5.10).

More than that we also have to guarantee, that machine i can only be chosen by user n if it is installed, that is only if for said machine i the binary variable y_i takes value 1 (5.11).

Constraints (5.12) and (5.13) reproduce the ratio between the probability of choosing machine i and the probability of choosing the opt-out alternative that is implied by the MNL model. Together, these constraints constitute the linear reformulation of the nonlinear MNL choice probabilities, acting like an upper and lower bound. Specifically, Constraint (5.12) defines an upper bound, while Constraint (5.13) defines a corresponding lower bound for the auxiliary variable, ensuring consistency between the MIP formulation and the original MNL model.

Finally two resource constraints are introduced: (5.14), where K is the maximum number of candidate locations to be chosen by the optimisation model, and (5.15) which prevents the fixed costs, summed up over all installed RVM locations, from exceeding the Budget B [Rogosinski et al., 2026].

6. Evaluation

This chapter reports the computational results of the deployment model presented in Chapter 5.

6.1 Computational Performance

The model was implemented in the academic version of GAMSPy and solved using the HiGHS (High-performance software for linear optimisation) solver.

All experiments were carried out in Python 3.12.10 on a computer running Windows 11, equipped with an Intel processor and 15.79 GB RAM. The solver successfully found the optimal solution with a runtime of 1.4930 seconds.

6.2 Optimal RVM Deployment

For the implementation a sample of 50 rows was randomly selected from the data set. All 35 candidate locations from the dataset were considered for the optimisation. The maximum number of deployable RVMs K was set to a value of 10 allowing the optimisation model to select at most 10 of the 35 candidate locations.

The fixed installation costs were assumed to amount to 24M COP per machine. Based on this assumption we multiplied the fixed cost by the total of 35 candidate locations, to get to a total Budget of 840M COP. The monetary value per gram of g_n is assumed to be 2.3 COP in our model implementation.

Since the optimisation was performed on a sample of only 50 randomly selected rows drawn from the total of 2,255 observations, the amount of returned plastic g_n was scaled by a factor of 10000, in order to represent a larger population and to ensure that revenues and investment costs are expressed on a comparable scale.

Table 6.1 shows the results of our optimisation model.

For an optimal RVM deployment the model chose machine 4, despite allowing the model to choose up to 10 machines. Moreover only 24M COP (which is the exact

Table 6.1: Summary of the optimisation results.

Category	Metric	Value
Economic	Net Objective	204,830,253 COP
	Revenue	228,830,253 COP
	Fixed Costs	24,000,000 COP
Decision	Selected Machine(s)	Machine 4
	Used K-Constraint	1 of 10
	Budget Used	24M / 840M COP
Customer Behaviour	Choice Probability (Site 4)	90–95%
	Opt-out Probability	5–6%

value of the fixed cost per RVM) out of a total Budget of 840M COP was used. The results suggest that only machine 4 is profitable enough to cover the fixed costs per machine of 24M COP. To test this hypothesis, a sensitivity analysis regarding the budget amount was conducted as follows.

6.3 Sensitivity Analysis

Since the baseline optimisation model selected only machine 4 a sensitivity analysis was done to test how the results change when varying the budget. Table 6.2 reports the selected machine and objective value for each budget level.

Table 6.2: Budget sensitivity analysis.

Budget (M COP)	Selected RVM	Objective
0	None	0.00
24	Machine 4	204.83
48	Machine 4	204.83
72	Machine 4	204.83
96	Machine 4	204.83
120	Machine 4	204.83
144	Machine 4	204.83
168	Machine 4	204.83
192	Machine 4	204.83
216	Machine 4	204.83
240	Machine 4	204.83

The available budget was varied from 0 to 240 million COP, as Figure 6.1 illustrates. The sensitivity analysis reveals a single breaking point at an available budget of 24M COP. Once the breakpoint is reached, the model chooses machine 4. Despite the growing budget, the model's results do not change. This result indicates that the budget constraint is binding only for budgets below 24 million COP. Above this value, additional budget has no marginal value for the location opening decisions, since for all other machines, the fixed costs are too high relative to the machine's

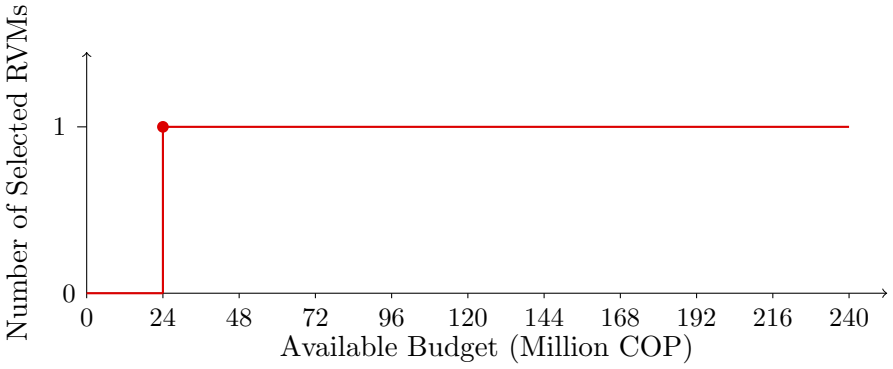


Figure 6.1: Number of selected RVMs as a function of the available budget. The breakpoint at 24 million COP indicates the minimum investment required to install the first RVM.

marginal utility. This implies that only machine 4 is economically profitable enough to generate enough profit to cover the fixed costs. The result also suggests that the limited number of selected RVMs is not caused by a limited budget, but rather by insufficient economical attractiveness of all machines, except machine 4. Hence, the limiting factor in the optimisation model is not the available investment budget but the economic attractiveness of the candidate RVM locations. It is therefore not enough to simply increase the budget. Rather, we must examine to what extent the candidate RVM locations can be made more accessible and appealing to customers from an economic perspective.

7. Conclusion

This paper applied a choice-based facility location framework to reverse vending machine deployment. It combined a discrete choice model of user behaviour with a mixed integer optimisation model. This chapter summarises how both components developed over the project and what the final results mean.

From Model 1 to the Final Specification

The discrete choice model went through three stages. Model 1 included all 35 candidate machines as available alternatives for every individual. This produced a strong distance effect, but rested on an unrealistic assumption: that every respondent considered all 35 machines. The choice set was therefore restricted to the 12 machines that actually appeared as observed choices, plus the opt-out alternative. On this restricted set, Model 2 included a standalone indoor coefficient and Model 3 removed it. A likelihood-ratio test ($LR = 0.022$, against a critical value of 3.841 at the 5% level) showed that the indoor coefficient did not improve fit significantly. Model 3 was retained as the final specification.

Behavioural Insights

In Model 3, building type has by far the largest effect on utility. Machines at private or public buildings are strongly preferred; machines in business or campus areas are avoided. The combined distance-indoor effect is 0.007, with a standard error of about 0.028 and a t-statistic of roughly 0.25 – not statistically different from zero. Indoor machines are, in practice, distance-independent. This is plausible if indoor placement coincides with buildings people already visit regularly. Older users avoid indoor machines somewhat more than younger users; female users show a mild preference for indoor over outdoor machines.

A Note on Generalisation

Model 3 relies on generic rather than alternative-specific coefficients (Chapter 4). This is what allows a model estimated on only 12 observed machines to be applied

to all 35 candidate locations in the optimisation step. It assumes that machines not observed during estimation affect utility purely through the same measured attributes, with no unobserved machine-specific factors. This assumption is necessary for the extrapolation to work, but it also means the optimisation results for previously unobserved sites are only as reliable as this assumption holds.

From Pure Coverage to a Monetised Optimisation Model

The optimisation model likewise developed in two stages. The base formulation, following Haase and Müller [2014] directly, maximises expected captured demand under a pure cardinality constraint, with no cost or budget consideration. For this project, the formulation was extended with a monetary objective and per-site fixed installation costs, together with a budget constraint. This extension goes beyond the literature reviewed in Section 2.1.

Optimisation Result

Applied to the final coefficients from Model 3, a sample of 50 users, and all 35 candidate sites, with $K = 10$ and a budget of 840 million COP, the model selects only Machine 4. It captures 90–95% of choice probability across users, with the opt-out alternative at 5–6%. Net profit is roughly 205 million COP, against a fixed cost of 24 million COP for the single installed machine – only 1 of 10 permitted sites, and only 24 of 840 million COP in budget, are used.

A budget sensitivity analysis, varying the available budget from 0 to 240 million COP, clarifies why. The model's decision does not change once the budget exceeds 24 million COP: no additional machine ever becomes worth installing, regardless of how much further budget is made available. This indicates that the limiting factor is not the size of the budget, but the economic attractiveness of the remaining candidate sites relative to their fixed cost. Machine 4 alone generates enough revenue to clear the 24-million-COP threshold; none of the others do, at any budget level tested.

This result is a direct, practical consequence of the IIA property discussed in Section 4.1. Because substitution between machines is proportional and does not depend on spatial proximity, one dominant site can absorb almost all reachable demand. The model has no mechanism to reward opening a second, nearby machine unless it is independently profitable – regardless of how much budget is available for it.

Compared to deterministic facility location approaches, such as the maximal covering model of Bedoya et al. [2024], the model used here has the conceptual advantage of representing explicitly how users choose among available machines. A direct numerical comparison is not meaningful, since the two approaches differ in objective, data, and demand assumptions, but the behavioural grounding of the choice-based approach is a clear conceptual improvement over models that treat demand as fixed.

Summary

Over the project, the discrete choice model moved from an unrealistic full-choice-set specification to a parsimonious model with generic, extrapolable coefficients. The optimisation model moved from a pure coverage objective to one that incorporates

cost and budget explicitly, and the resulting sensitivity analysis showed that budget was never the binding constraint at all. A single, well-placed machine captures nearly all reachable demand; further expansion is limited by the underlying behavioural model itself, not by available investment.

7.1 Future Steps

The results point to three directions for extending this work.

Continuous Relaxation

The binary decision variable y_i forces an all-or-nothing choice per candidate site. The sensitivity analysis in Section 6.3 shows why this matters here: no site other than Machine 4 is profitable enough to justify a full installation, but this does not necessarily mean these sites have zero potential. A continuous relaxation, replacing y_i with a variable r_i representing partial capacity or accessibility, could represent intermediate options – for instance, a smaller-format installation, or shared capacity between nearby low-demand sites – that the current binary formulation cannot express.

A first-order stationarity condition for such a continuous relaxation was outlined in discussion with the project supervisor. Its formal derivation was not pursued as part of this work, but it conceptually mirrors the same marginal trade-off underlying the discrete result: each additional unit of accessibility at site i competes with the accessibility already captured by other sites. Formalising this relaxation connects to the outer-approximation methods reviewed in Section 2.1 [Ljubić and Moreno, 2018; Mai and Lodi, 2020], which handle related continuous extensions of the discrete facility location problem.

Relaxing IIA

Section 2.2 discussed Rogosinski et al. [2026], who show that relaxing the IIA property through a mixed logit model can substantially change optimisation outcomes when substitution between alternatives is correlated rather than proportional. This is directly relevant here: the result that a single machine absorbs nearly all demand is, in part, a consequence of the MNL's IIA assumption, which cannot represent that some machines may be closer substitutes for each other than others. A mixed logit specification, allowing correlated taste parameters across nearby or similar machines, could reveal whether the observed concentration on Machine 4 is a genuine behavioural pattern or partly an artefact of the model's substitution structure.

Dynamic and Multi-Objective Extensions

The current model is static: it optimises a single snapshot of demand and cost. Demand for RVMs may change over time as usage habits or local infrastructure develop, which a dynamic formulation could capture. Similarly, the current objective is purely economic. A planner interested in geographic coverage or equitable access, not only profit, might prefer a more distributed deployment even where doing so is not strictly profit-maximising – a trade-off explicitly addressed in multiobjective RVM models such as Bedoya et al. [2024], though without a behavioural demand component. Combining their multiobjective structure with the choice-based demand model of this project is a natural extension.

Remarks on the Use of AI Tools

The authors used the AI assistant Claude (Anthropic) while preparing this seminar paper. It supported the assembly of the individually written chapters into a single LaTeX document, the formatting of tables, equations, and one figure, the verification of the bibliography, and the unification of language and terminology. The scientific content, that is the discrete choice and optimisation models, their estimation and implementation, the computational results, and the review and interpretation of the literature, was developed by the authors. Every AI-assisted passage was reviewed and, where necessary, revised by the authors, who take full responsibility for the final content. No AI tool was used to generate or alter data or computational results.

Bibliography

S. Bedoya, J. L. Mejía, P. Maya-Duque, and J. G. Villegas. A Multiobjective Maximal Covering/p-Dispersion Model for Reverse Vending Machine Location. In *Evolution and Trends of Sustainable Approaches*, pages 111–127. Elsevier, 2024. doi: 10.1016/B978-0-443-21651-0.00015-2. (cited on Page 4, 5, 21, and 22)

Stefano Benati and Pierre Hansen. The Maximum Capture Problem with Random Utilities: Problem Formulation and Algorithms. *European Journal of Operational Research*, 143(3):518–530, 2002. doi: 10.1016/S0377-2217(01)00340-X. (cited on Page 2 and 3)

Michel Bierlaire. A Short Introduction to Biogeme. Technical Report TRANSP-OR 230620, Transport and Mobility Laboratory, École Polytechnique Fédérale de Lausanne, 2023. (cited on Page 9)

Rubén Cordera, Esther González-González, Soledad Nogués, Julián Arellana, and José L. Moura. Modal Choice for the Driverless City: Scenario Simulation Based on a Stated Preference Survey. *Journal of Advanced Transportation*, 2022:1–12, 2022. doi: 10.1155/2022/1108272. (cited on Page 12)

Maria Eduarda Fernandes and João Maurício C. Reverse vending machines in the circular economy: A bibliometric analysis of an underexplored enabler (2003–2023). *Cleaner Waste Systems*, 12:100404, 2025. doi: 10.1016/j.clwas.2025.100404. (cited on Page 2)

Martin Geissdoerfer, Paulo Savaget, Nancy M. P. Bocken, and Erik Jan Hultink. The Circular Economy – A new sustainability paradigm? *Journal of Cleaner Production*, 143:757–768, 2017. doi: 10.1016/j.jclepro.2016.12.048. (cited on Page 1)

Roland Geyer, Jenna R. Jambeck, and Kara Lavender Law. Production, use, and fate of all plastics ever made. *Science Advances*, 3(7):e1700782, 2017. doi: 10.1126/sciadv.1700782. (cited on Page 1)

Carlos Gomes da Silva, Maria Eduarda Fernandes, and João Maurício C. Sustainable Location of Reverse Vending Machines. *Circular Economy and Sustainability*, 6 (2):106, 2026. doi: 10.1007/s43615-026-00825-2. (cited on Page 4 and 5)

Knut Haase and Sven Müller. A Comparison of Linear Reformulations for Multinomial Logit Choice Probabilities in Facility Location Models. *European Journal of Operational Research*, 232(3):689–691, 2014. doi: 10.1016/j.ejor.2013.08.009. (cited on Page 2, 3, 4, 5, 15, and 21)

David L. Huff. Determination of Intra-Urban Retail Trade Areas. Technical report, Real Estate Research Program, University of California, Los Angeles, 1962. (cited on Page 4)

David L. Huff. Defining and Estimating a Trading Area. *Journal of Marketing*, 28 (3):34–38, 1964. doi: 10.1177/002224296402800307. (cited on Page 4)

Frank S. Koppelman and Chandra R. Bhat. A Self-Instructing Course in Mode Choice Modeling: Multinomial and Nested Logit Models. Technical report, U.S. Department of Transportation, Federal Transit Administration, 2006. (cited on Page 12)

Ralf Krohn, Sven Müller, and Knut Haase. Preventive Healthcare Facility Location Planning with Quality-Conscious Clients. *OR Spectrum*, 43:59–87, 2021. doi: 10.1007/s00291-020-00605-w. (cited on Page 14)

Yun Hui Lin, Yuan Wang, Dongdong He, and Loo Hay Lee. Last-Mile Delivery: Optimal Locker Location under Multinomial Logit Choice Model. *Transportation Research Part E: Logistics and Transportation Review*, 142:102059, 2020. doi: 10.1016/j.tre.2020.102059. (cited on Page 4 and 5)

Ivana Ljubić and Eduardo Moreno. Outer Approximation and Submodular Cuts for Maximum Capture Facility Location Problems with Random Utilities. *European Journal of Operational Research*, 266(1):46–56, 2018. doi: 10.1016/j.ejor.2017.09.023. (cited on Page 3 and 22)

Tien Mai and Andrea Lodi. A Multicut Outer-Approximation Approach for Competitive Facility Location under Random Utilities. *European Journal of Operational Research*, 284(3):874–881, 2020. doi: 10.1016/j.ejor.2020.01.020. (cited on Page 3 and 22)

Daniel McFadden. Conditional Logit Analysis of Qualitative Choice Behavior. In Paul Zarembka, editor, *Frontiers in Econometrics*, pages 105–142. Academic Press, New York, 1974. (cited on Page 9)

OECD. Extended Producer Responsibility: Updated Guidance for Efficient Waste Management. Technical report, OECD Publishing, Paris, 2016. (cited on Page 1)

OECD. Global Plastics Outlook: Economic Drivers, Environmental Impacts and Policy Options. Technical report, OECD Publishing, Paris, 2022. (cited on Page 1)

Stefan Rogosinski, Sven Müller, and Lorena Reyes-Rubiano. Distribution-Specific Approximation Guarantees for the Random-Parameters Logit Assortment Problem. *European Journal of Operational Research*, 333:216–233, 2026. doi: 10.1016/j.ejor.2025.11.016. (cited on Page 4, 9, 16, and 22)

Matthew S. Savoca, Erin L. Murphy, Anja M. Brandon, Leah M. Thornton Hampton, Lauren Roman, Majel Kong, Maria Eugenia Fernandes, Sika Abrokwah, Cecelia Darter, François Galgani, Jennifer F. Provencher, Chibuisi Alimba, Robson G. Santos, and Britta Denise Hardesty. The effects of plastic pollution on biodiversity. *Nature Reviews Biodiversity*, 2(7):425–440, 2026. doi: 10.1038/s44358-026-00170-0. (cited on Page 1)

Jason A. Somarelli, Jason W. Arnold, and Andrew B. West. The carcinogenic consequences of the plastic pollution crisis. *The Journal of Clinical Investigation*, 136(4):e203775, 2026. doi: 10.1172/JCI203775. (cited on Page 1)

Kenneth E. Train. *Discrete Choice Methods with Simulation*. Cambridge University Press, 2 edition, 2009. (cited on Page 8, 9, 10, and 11)

I herewith assure that I wrote the present thesis independently, that the thesis has not been partially or fully submitted as graded academic work and that I have used no other means than the ones indicated. I have indicated all parts of the work in which sources are used according to their wording or to their meaning.

Aachen, July 27th, 2026